

Jatin Arutla

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EDUCATION

University of Birmingham

Master of Science in Artificial Intelligence and Machine Learning | Grade: Distinction

Birmingham, UK

2022 – 2023

- Advisor: Dr Jianbo Jiao

Gandhi Institute of Technology and Management (GITAM) University

Bachelor of Technology in Computer Science and Engineering | GPA: 9.08/10 | 3rd out of 60

Visakhapatnam, India

2017 – 2021

- Advisor: Dr Srinivasa Chakravarthy Lade

RESEARCH EXPERIENCE

Reference Shift in EEG Decoders and Pretrained Encoders

Independent Research (Ongoing)

- Identified EEG referencing as a source of train-test distribution shift, with substantial performance drops when models were evaluated under a different reference or montage from training.
- Built train-versus-test transfer matrices across seven referencing operators using CSP-LDA, ShallowConvNet, REVE and MIREPNet, and tested robustness to held-out references and operator families.
- Showed that CAR removes much of the mismatch between global references, while Euclidean Alignment reduces the broader transfer gap to around 2–4 percentage points.
- Tested limited-target alignment and found that even a small amount of unlabeled target data can sharply reduce the reference-specific mismatch.

Layer-Resolved Speech and Text Model Encoding of ECoG Responses

GitHub

Independent Research

- Rebuilt a layer-resolved encoding analysis on the Podcast ECoG dataset, using banded ridge regression and variance partitioning to compare Whisper representations with acoustic, phonetic and syntactic features.
- Extended the analysis to Qwen2.5 transcript representations to compare speech and text model representations and examine how their relationship with neural responses changed across model depth.
- Found that neural prediction and unique variance explained by Whisper increased across model depth, while later representations were less explained by interpretable speech features. Whisper also showed substantially stronger acoustic overlap than Qwen.

Weakly Supervised Lung Disease Localization with Self-Supervised Pretraining

Thesis

Advisor: Dr Jianbo Jiao

- Developed a mask-guided Weakly Supervised Object Localization (WSOL) framework on the NIH ChestX-ray14 dataset, using LayerCAM-based disease masks across all layers to improve localization without bounding boxes.
- Introduced a modified perceptual loss with shallow-layer features for image alignment and an adaptive Otsu thresholding method for bounding box extraction from Class Activation Maps.
- Evaluated self-supervised, supervised, and random initializations, achieving +8.2% mean IoU over baseline and improving weakly supervised localization across the evaluated thoracic disease classes.

Prognostication of Unseen Objects using Zero-shot Learning

Paper

Advisor: Dr Srinivasa Chakravarthy Lade

- Developed a zero-shot learning framework combining VGG16 visual embeddings with Word2Vec semantic vectors to recognize unseen food categories.
- Benchmarked zero-shot learning against fully supervised approaches, tuning model hyperparameters and evaluating transfer performance.
- Achieved 92% Top-5 accuracy on unseen classes, demonstrating the feasibility of semantic feature transfer for image recognition without labeled examples.

PUBLICATION

Srinivasa L. Chakravarthy and **Jatin V.R. Arutla**. “Prognostication of Unseen Objects using Zero-Shot Learning with a Complete Case Analysis.” Interdisciplinary Description of Complex Systems: INDECS, Vol. 20, No. 4, 2022.

SELECTED TRAINING

Computational Neuroscience

2026

Neuromatch

- Completed an intensive 3-week program focusing on computational neuroscience and neural data modeling.
- Built mechanistic and descriptive models in Python, covering generalized linear models, dynamical systems, reinforcement learning and causal analysis of neural activity.
- Completed a group ECoG project decoding faces and houses under visual noise, examining how neural classification changed across noise levels, time and brain regions.

AWARDS

- Commonwealth Games Scholarship — University of Birmingham 2022
- Merit Scholarship for Academic Excellence — GITAM University 2017–2021

TECHNICAL SKILLS

Core: Python (PyTorch, TensorFlow, TorchAudio, Braindecode, MNE-Python)

ML & Stats: NumPy, SciPy, scikit-learn, Pandas, Matplotlib

Tools: Git, Streamlit

Other: SQL, MATLAB

WORK EXPERIENCE

Data Scientist

Dec. 2023 – Present

Fonz Leather Styles Limited (Ashwood Leather)

Birmingham, UK/ Remote

- Delivered a Python and Streamlit analytics app unifying sales, marketing, inventory, and ads data for a site with 100k monthly visitors. Cut reporting latency from about 24 hours to seconds.
- Developed and executed data-driven advertising strategies across Google, Amazon, and Debenhams, using performance analytics and dynamic budget allocation to achieve a 40% improvement in ROAS.
- Built a demand forecasting system using time-series models (Prophet, XGBoost), cutting stockouts and reducing excess inventory by 20% through earlier replenishment signals.
- Designed a segmentation pipeline for 80k+ customers with K-Means and RFM analysis, producing audience segments for targeted email campaigns. Delivered a 25% lift in targeted campaign sales versus broad sends.

Data Science Intern

July 2021 – Sep. 2021 & Feb. 2022 – June 2022

Evolv Fit

New Delhi, India

- Developed a real-time pose detection system using Google MediaPipe and integrated it into a React Native app with TensorFlow and OpenCV, enabling accurate tracking of exercise form and rep counting.
- Designed a personalized recipe recommendation system using NLP, tailored to fitness goals and dietary preferences, increasing user engagement through customized meal suggestions.

PROJECTS

Cross-Subject Motor-Imagery EEG Decoding

GitHub

- Extended ATCNet for four-class motor-imagery decoding on BCI Competition IV-2a with Euclidean Alignment and SimCLR-style contrastive pretraining, evaluated under subject-dependent and leave-one-subject-out protocols.
- Achieved 79.2% mean accuracy with aligned features and released a reproducible pipeline for EEG preprocessing, augmentation, training, and cross-subject benchmarking.

2-D and 3-D Brain MRI Tissue Segmentation

GitHub

- Implemented Otsu's method and Gaussian Mixture Model-based Hidden Markov Random Field algorithms in MATLAB for MRI tissue segmentation.
- Achieved Dice coefficients of 0.75 (2-D) and 0.80 (3-D), demonstrating the potential of probabilistic methods for biomedical imaging.

Aspect-based Sentiment Analysis of Tech Product Reviews

GitHub

- Built a multi-label sentiment analysis pipeline linking Entity–Attribute–Polarity tags with POS features, bag-of-words, and text augmentation (synonyms, back-translation) to extract fine-grained insights from product reviews.
- Preprocessed multi-class, multi-label review data at sentence and document levels and trained Logistic Regression models for reliable opinion classification and feature-level sentiment detection.